

# JEA (JAY) KWON

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## RESEARCH INTERESTS

- Machine memory and AI engrams: localizing, editing, and erasing learned knowledge in LLM parameters
  - Mechanistic interpretability, steerability, and alignment of large language models
  - Neuro4AI and AI4Neuro: brain-inspired AI and AI-driven neuroscience
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## EDUCATION

- Korea University, Seoul, South Korea • Ph.D. in Nano-Bio-Information-Technology • 2014 - 2022  
Advisor: C. Justin Lee, Director of IBS
  - Saitama University, Saitama, Japan • B.E. in Electrical and Electronic System • 2008 - 2012
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## PROFESSIONAL EXPERIENCE

- Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany • Postdoctoral Researcher • 2024 - Present  
Advisor: Meeyoung Cha, Director of MPI-SP
  - Center for Cognition and Sociality, Institute for Basic Science (IBS), Daejeon, South Korea • Postdoctoral Researcher • 2022 - 2024
  - Center for Cognition and Sociality, Institute for Basic Science (IBS), Daejeon, South Korea • Researcher • 2018 - 2020
  - Korea Institute of Science and Technology (KIST), Seoul, South Korea • Student Researcher • 2014 - 2018
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## ACHIEVEMENTS

- European Commission, Horizon Europe (HORIZON-MSCA-2025-PF-01-01) • Seal of Excellence — MSCA Postdoctoral Fellowships 2025 • 2026  
Proposal 101286042 — TRACE: Towards Reliable AI through Cognitive Engrams (with Max-Planck-Gesellschaft), total score 89.40%
- Korean Artificial Intelligence Association (JKAIA) • Excellent Paper Award • 2023  
Brain-inspired Lp-Convolution benefits large kernels and aligns better with visual cortex
- Application Number: 10-2023-0027219 • Patent Application & Technology Transfer • 2023  
Device and Method for Classification of Human and Animal Behavior, Patent Contribution=35%  
Finalized a technology transfer with ACTNOVA Inc.
- Korean Society for Brain and Neural Sciences (KSBNS) • Best Presentation Award • 2022  
ABCD-analysis: Mapping animal behavior and differential analysis of kinematic features without selection bias

- Korean Society for Molecular and Cellular Biology (KSMCB) • Young Investigator Award • 2022  
Retina-attached slice recording reveals light-triggered tonic GABA signaling in suprachiasmatic nucleus
- Center for Cognition and Sociality (CCS) Workshop • Best Poster Award • 2021  
Mapping Behavior with DECLARE: Deep Embedded Clustering of Action REpresentation
- Korea University - Korea Institute of Science and Technology (KU-KIST) Graduate School • Full-Ride Scholarship for Doctoral Studies • 2014 - 2022
- Korea-Japan Governments Joint Scholarship Program for Science and Engineering Students • Full-Ride Scholarship for Bachelor's Studies • 2007 - 2012

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#### ORGANIZING EXPERIENCE

- Korean Society for Brain and Neural Sciences (KSBNS) Annual Meeting • Symposium Organizer: "Minds Meet Machines: Bidirectional Innovations in AI and Neuroscience" • 2026
- KSBNS & CJK (China-Japan-Korea) Joint Symposium • Symposium Co-Organizer: "Neuroscience-inspired AI: Computational Insights into Biological and Artificial Intelligence" • 2025
- KAIST-NeuroAI Journal Club • Founding Member • 2020

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#### PROGRAM COMMITTEES & REVIEWING

- AAAI 2026, AI Alignment Track • Program Committee • 2026
- Machine+Behavior Conference 2026, Harnack House, Berlin • Program Committee • 2026
- International Conference on Learning Representations (ICLR) • Reviewer • 2026
- ICLR Re-Align Workshop • Reviewer • 2025 - 2026

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#### INVITED TALKS

- Scalable Trustworthy AI (STAI) group - KAIST • AI Engram: In Search of Memory Traces in Artificial Neural Networks • 2026
- Introduction to System Health - Ewha Womans University • AI and Neuroscience • 2026
- Max Planck Institute for Software Systems • AI Engram: In Search of Memory Traces in Artificial Neural Networks • 2026
- Spark(l)ing Science - Max Planck Institute for Security and Privacy • AI Engram: In Search of Memory Traces in Artificial Neural Networks • 2026
- KAIST BK21 IBS Symposium - Integrated Neuroscience and Physiology • How neuroscience can benefit AI • 2025
- KSBNS Symposium • Transformer as a Hippocampal Memory Consolidation Model based on NMDAR-inspired Nonlinearity • 2025
- Netzwerk Junge Generation Deutschland-Korea • Sovereign AI: K-AI as an opportunity • 2025
- Institute for Basic Science (IBS) • Brain-inspired AI • 2025

- Korea Brain Research Institute (KBRI) • Brain-inspired AI • 2025
- Gwangju Institute of Science and Technology (GIST) • Brain-inspired AI • 2025
- MLAI group, Yonsei University • Neuroscience for Artificial Intelligence • 2025
- Kyungpook National University (KNU) • Brain-inspired AI • 2024
- Korea Institute of Ocean Science & Technology (KIOST) • SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior • 2024
- Korea Brain Research Institute (KBRI) • SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior • 2024
- IBS-CCS Symposium • Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity • 2023
- IBS Science in the Cinema Roof • Memory editing in the brain - Eternal Sunshine of the Spotless Mind • 2023
- Reading Festival at Jeonil High School • From human mind to artificial intelligence - Is ChatGPT a bridge • 2023
- IBS Data Science Group seminar • Resolving clustering bias during animal behavior mapping and kinematic profiling • 2022
- IBS 2021 2nd Workshop • Age-dependent motor coordination analysis by 3D pose-estimation using AVATAR • 2021
- KAIST-NeuroAI Journal Club • Backpropagation and the Brain • 2020
- IBS 2019 Year-End Workshop • Deep learning with NMDA receptor inspired activation function • 2020
- Korean Society for Molecular and Cellular Biology (KSMCB) • Development of a Low-cost, Comprehensive Recording System for Circadian Rhythm Behavior • 2019

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## PUBLICATIONS

37. **J. Kwon\***, D. Kim\*, J. Kim\*, Y. Kim, W. Kook, M. Cha.  
AI Engram: In Search of Memory Traces in Artificial Intelligence.  
*International Conference on Machine Learning (Spotlight, top 2.2%)*, 2026.
36. M. S. Locatelli, F. Tonucci, **J. Kwon**, L. F. Vecchiotti, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha.  
Textual Supervision Enhances Geospatial Representations in Vision-Language Models.  
*International Conference on Machine Learning*, 2026.
35. J. Kim, **J. Kwon**, L. F. Vecchiotti, W. Dong, J. Kim, M. Cha.  
Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions.  
*Findings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026.
34. M. Kim, D. Kim, **J. Kwon**, N. Yang, K. Jung, M. Cha.  
How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models.  
*Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026.

33. S. Kim, **J. Kwon**.  
From Human Cognition to AI Reasoning: Models, Methods, and Applications.  
*ICLR HCAIR Workshop*, 2026.
32. **J. Kwon\***, S. Kim\*, J. Woo, K. Tanaka-Yamamoto, O. James, E. De Schutter, S. Hong, C. J. Lee.  
Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination.  
*Experimental & Molecular Medicine*, 2026.
31. **J. Kwon\***, L. F. Vecchietti, S. Park, M. Cha.  
Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment.  
*AAAI Conference on Artificial Intelligence*, 2026.
30. D. Kim, M. Kim, **J. Kwon**, N. Yang, M. Cha.  
Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing.  
*International Conference on Learning Representations*, 2026.
29. N. Yang, D. Kim, **J. Kwon**, M. Kim, K. Jung, M. Cha.  
Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning.  
*International Conference on Learning Representations*, 2026.
28. M. Kim, **J. Kwon**, D. Kim, M. Cha.  
In Search of the Engram in LLMs: A Neuroscience Perspective on the Memory Functions in AI Models.  
*The Fourth Blogpost Track at ICLR 2025*, 2025.
27. B. Hyeon, J. Shin, J. Lee, W. Kim, **J. Kwon**, H. Lee, D. Kim, C. Y. Kim, S. Choi, J. Jeong, K. Kim, C. J. Lee, D. Kim, W. D. Heo.  
Integrating Artificial Intelligence and Optogenetics for Parkinson's Disease Diagnosis and Therapeutics in Male Mice.  
*Nature Communications*, 2025.
26. J. Kim\*, **J. Kwon\***, L. F. Vecchietti\*, A. Oh, M. Cha.  
Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment.  
*ICLR 2025 Workshop in Bi-Align*, 2025.
25. **J. Kwon\***, S. Lim, K. Song, C. J. Lee.  
Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex.  
*International Conference on Learning Representations*, 2025.
24. **J. Kwon\***, M. Sa, H. Kim, Y. Seong, C. J. Lee.  
Egocentric 3D Skeleton Learning in Identity-Aware Deep LSTM Network Encodes Obese-Like Motion Representations.  
*ICLR TS4H Workshop*, 2024.
23. **J. Kwon\***, S. Lim, K. Song, C. J. Lee.  
Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex.  
*ICLR Re-Align Workshop*, 2024.
22. **J. Kwon\***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee.  
SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior.  
*International Journal of Computer Vision*, 2024.
21. **J. Kwon\***, M. Sa, H. Kim, Y. Seong, C. J. Lee.  
Egocentric 3D Skeleton Learning in a Deep Neural Network Encodes Obese-like Motion Representations.  
*Experimental Neurobiology*, 2024.

20. H. Kang, A. Han, A. Zhang, H. Jeong, W. Koh, J. M. Lee, H. Lee, H. Y. Jo, M. A. Maria-Solano, M. Bhalla, **J. Kwon**, et al..  
GolpHCat (TMEM87A), a Unique Voltage-Dependent Cation Channel in Golgi Apparatus, Contributes to Golgi-pH Maintenance and Hippocampus-Dependent Memory.  
*Nature Communications*, 2024.
19. D. Kim\*, **J. Kwon\***, M. Cha, C. J. Lee.  
Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity.  
*Advances in Neural Information Processing Systems*, 2023.
18. M. Sa, E. Yoo, W. Koh, M. G. Park, H. Jang, Y. R. Yang, M. Bhalla, J. Lee, J. Lim, W. Won, **J. Kwon**, et al..  
Hypothalamic GABRA5-Positive Neurons Control Obesity via Astrocytic GABA.  
*Nature Metabolism*, 2023.
17. **J. Kwon\***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee.  
SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior.  
*CVPR CV4Animals Workshop*, 2023.
16. M. Nam, H. Y. Ko, D. Kim, S. Lee, Y. M. Park, S. J. Hyeon, W. Won, J. Chung, S. Y. Kim, H. H. Jo, K. T. Oh, Y. Han, G. Lee, Y. H. Ju, H. Lee, H. Kim, J. Heo, M. Bhalla, K. J. Kim, **J. Kwon**, et al..  
Visualizing Reactive Astrocyte-Neuron Interaction in Alzheimer's Disease Using 11C-Acetate and 18F-FDG.  
*Brain*, 2023.
15. D. Kim, **J. Kwon**, M. Cha, C. J. Lee.  
Transformer Needs NMDA Receptor Nonlinearity for Long-Term Memory.  
*NeurIPS Memory in Artificial and Real Intelligence Workshop*, 2022.
14. D. Kim, J. Kim, W. Jung, J. Park, M. Kim, A. Shin, Y. Jeong, S. Park, G. Shin, Y. W. Lee, **J. Kwon**, D. Kim.  
AVATAR: AI Vision Analysis for Three-Dimensional Action in Real-Time.  
*CVPR CV4Animals Workshop*, 2022.
13. S. Kim, **J. Kwon**, M. G. Park, C. J. Lee.  
Dopamine-Induced Astrocytic Ca<sup>2+</sup> Signaling in mPFC is Mediated by MAO-B in Young Mice, but by Dopamine Receptors in Adult Mice.  
*Molecular Brain*, 2022.
12. Y. H. Ju, M. Bhalla, S. J. Hyeon, J. E. Oh, S. Yoo, U. Chae, **J. Kwon**, W. Koh, J. Lim, Y. M. Park, et al..  
Astrocytic Urea Cycle Detoxifies A $\beta$ -Derived Ammonia While Impairing Memory in Alzheimer's Disease.  
*Cell Metabolism*, 2022.
11. J. M. Lee, M. Sa, H. An, J. M. J. Kim, **J. Kwon**, B. Yoon, C. J. Lee.  
Generation of Astrocyte-Specific MAOB Conditional Knockout Mouse with Minimal Tonic GABA Inhibition.  
*Experimental Neurobiology*, 2022.
10. **J. Kwon\***, M. W. Jang, C. J. Lee.  
Retina-Attached Slice Recording Reveals Light-Triggered Tonic GABA Signaling in Suprachiasmatic Nucleus.  
*Molecular Brain*, 2021.

9. M. W. Jang, T. Y. Kim, K. Sharma, **J. Kwon**, E. Yi, C. J. Lee.  
A Deafness Associated Protein TMEM43 Interacts with KCNK3 (TASK-1) Two-Pore Domain K<sup>+</sup> (K2P) Channel in the Cochlea.  
*Experimental Neurobiology*, 2021.
8. J. Won, H. H. Kazan, **J. Kwon**, M. Park, M. A. Ergun, S. Ozcan, B. Y. Choi, W. D. Heo, C. J. Lee.  
Ultimate COVID-19 Detection Protocol Based on Saliva Sampling and qRT-PCR with Risk Probability Assessment.  
*Experimental Neurobiology*, 2021.
7. K. Han, M. Lee, H. Lim, M. W. Jang, **J. Kwon**, C. J. Lee, S. Kim, M. Suh.  
Excitation-Inhibition Imbalance Leads to Alteration of Neuronal Coherence and Neurovascular Coupling Under Acute Stress.  
*Journal of Neuroscience*, 2020.
6. S. Oh, J. M. Lee, H. Kim, J. Lee, S. Han, J. Y. Bae, G. Hong, W. Koh, **J. Kwon**, E. Hwang, et al..  
Ultrasonic Neuromodulation via Astrocytic TRPA1.  
*Current Biology*, 2019.
5. Y. Han, **J. Kwon**, J. Won, H. An, M. W. Jang, J. Woo, J. S. Lee, M. G. Park, B. Yoon, S. E. Lee, et al..  
Tweety-Homolog (Ttyh) Family Encodes the Pore-Forming Subunits of the Swelling-Dependent Volume-Regulated Anion Channel (VRAC<sub>swell</sub>) in the Brain.  
*Experimental Neurobiology*, 2019.
4. J. Woo, J. O. Min, D. Kang, Y. S. Kim, G. H. Jung, H. J. Park, S. Kim, H. An, **J. Kwon**, J. Kim, et al..  
Control of Motor Coordination by Astrocytic Tonic GABA Release Through Modulation of Excitation/Inhibition Balance in Cerebellum.  
*Proceedings of the National Academy of Sciences*, 2018.
3. **J. Kwon\***, M. G. Park, S. E. Lee, C. J. Lee.  
Development of a Low-Cost, Comprehensive Recording System for Circadian Rhythm Behavior.  
*Experimental Neurobiology*, 2018.
2. **J. Kwon\***, H. An, M. Sa, J. Won, J. I. Shin, C. J. Lee.  
Orai1 and Orai3 in Combination with Stim1 Mediate the Majority of Store-Operated Calcium Entry in Astrocytes.  
*Experimental Neurobiology*, 2017.
1. G. E. Ha, J. Lee, H. Kwak, K. Song, **J. Kwon**, S. Jung, J. Hong, G. Chang, E. M. Hwang, H. Shin, et al..  
The Ca<sup>2+</sup>-Activated Chloride Channel Anoctamin-2 Mediates Spike-Frequency Adaptation and Regulates Sensory Transmission in Thalamocortical Neurons.  
*Nature Communications*, 2016.